EE4-05
EE9-CS7-22
EE9-FPN2-01

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

TRAFFIC THEORY & QUEUEING SYSTEMS

Monday, 26 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Special information for students

1. Erlang Loss formula recursive evaluation:

$$E_N(\rho) = \frac{\rho E_{N-1}(\rho)}{N + \rho E_{N-1}(\rho)}$$
$$E_0(\rho) = 1$$

2. Engset Loss formula recursive evaluation (for a fixed M and $p = \alpha/1+\alpha$):

$$e_N = \frac{(M - N + 1)\alpha e_{N-1}}{N + (M - N + 1)\alpha e_{N-1}}$$
$$e_0 = 1$$
$$\alpha = \lambda/\mu$$

3. Traffic capacity on basis of Erlang B formula (next page).

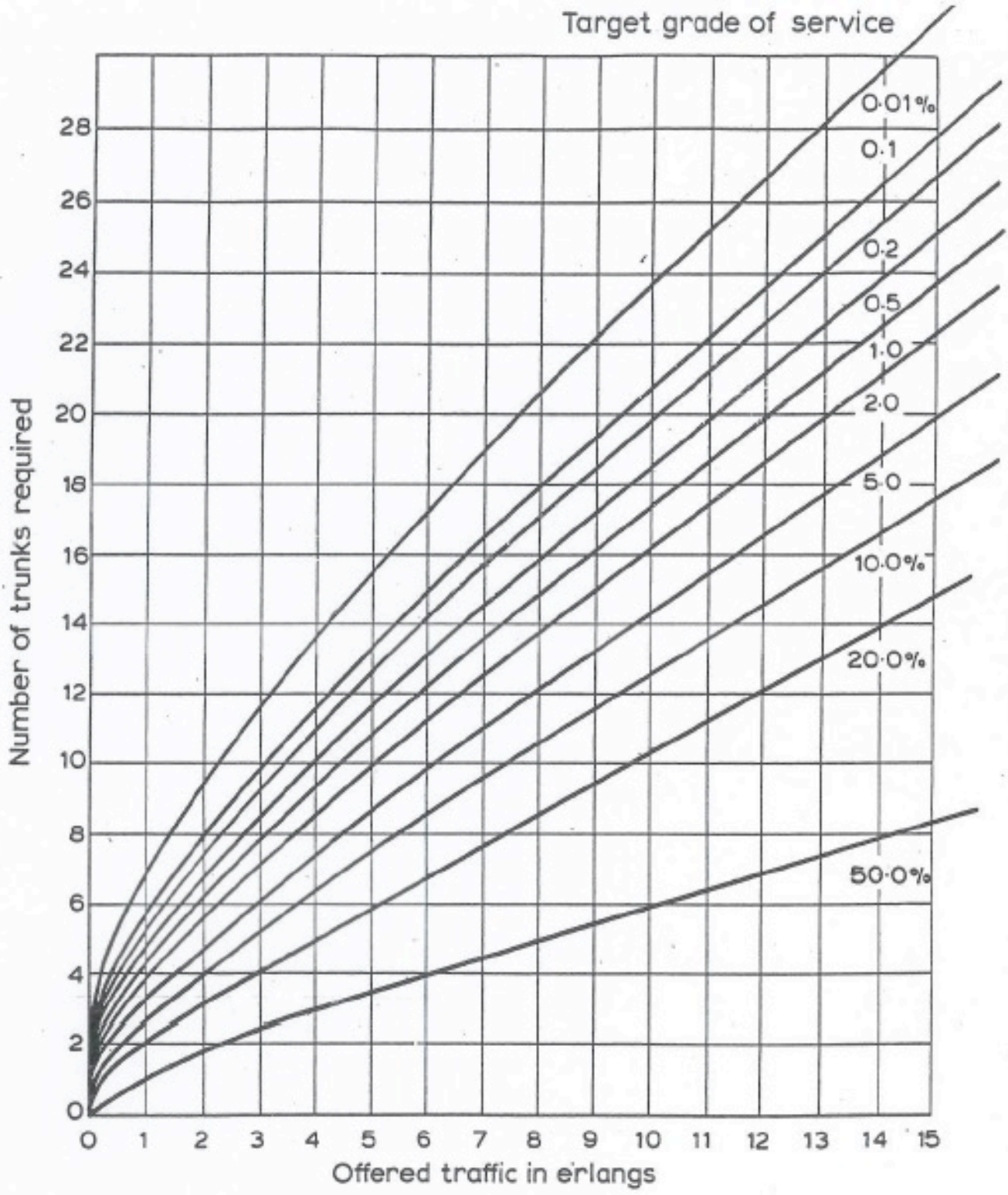
4. Expected residual time

$$E[R] = \frac{1}{2}\lambda E[S^2] = \frac{1}{2}\sum_{k=1}^{m}\lambda_k E[S_k^2]$$

5. Mean and variance of the Kosten overflow model (infinite overflow link $N = \infty$).

$$m = E[N_t] = \rho E_M(\rho)$$

$$v = Var(N_t) = m\left(1 - m + \frac{\rho}{M+m+1-\rho}\right)$$



Traffic capacity on basis of Erlang B. formula.

The Questions

1.

a) Pure chance traffic is offered to a 20-channel communication link.

Assume:

The holding time of any call is 120 seconds.

The arrival rate of call to the link is 300 calls per hour.

i) Estimate the mean carried traffic. [3]

ii) Estimate the variance of the carried traffic. [4]

iii) Estimate the mean channel occupancy. [3]

b) Consider the stream of requests from M identical and independent sources which are offered to a $N - channel$ link.

Each source, when idle, acts as a Poisson source.

i) Identify and discuss a suitable source model to carry out this analysis. [2]

ii) Derive the link state distribution for $M \leq N$. [3]

iii) Derive the link state distribution for $M > N$. [3]

iii) If the number of sources increases to a point where $M \gg N$, explain how the derivations in i) and ii) can be affected or simplified [2]

2.

a) In a call centre system, all arriving requests are accepted to a buffer which operates a FIFO discipline and is endowed with a group of 14 phone bots.

The traffic offered to this centre has been estimated at 11.2 erlangs and on average 288 calls are being received per hour.

Hint: Assume that the buffer is quite large, that is, no incoming request will be denied service.

i) Determine the probability that an incoming request will have to wait in the buffer before being answered by a phone bot. [5]

ii) Determine how long (on average) a request, which upon arrival is being placed in the buffer, will have to wait for its turn. [5]

b) For the call centre described in a), assume that in case of a failure, the call centre can still operate but with a finite buffer of size 10.

i) Derive the mean queue length seen by arriving requests that join the buffer. [4]

ii) Derive the average waiting time for an incoming request that has been accepted, but is being placed into the buffer upon arrival. [4]

iii) Discuss if the results obtained in 2a) ii) and 2b) ii) are as expected. [2]

3.

a)

i) Discuss in your own words how an $M/G/1$ queuing system could be used to analyse a pre-emptive restart priority scheme. [3]

ii) Derive a lower bound approximation to the transit time of the system described in i). [4]

b)

The two dimensional Birth and Death process shown in Fig. 3.1 represents the traffic being offered to an overflow link.

Demands to the first choice link arrive at a rate of 180 calls per hour and the mean call duration is 180 seconds.

The state space is defined by $S = (N_t, Y_t)$.

i) Describe what do the parameters N_t and Y_t represent.

How many overflow links is the model in Fig. 3.1. representing?

Assume that the following values are given: $\alpha = 0.2\, m^{-1}$ and $\beta = 7.0\, m^{-1}$.
Complete the transition rates missing from Fig. 3.1. [3]

ii) Derive the mean and variance of the overflow traffic. Clearly state all the steps in your derivation. [4]

iii) Derive the Call congestion (probability that the overflow channel is busy) and Time congestion (probability that an arrival finds the overflow channel busy). [4]

iv) Discuss what type of traffic is expected from this model. [2]

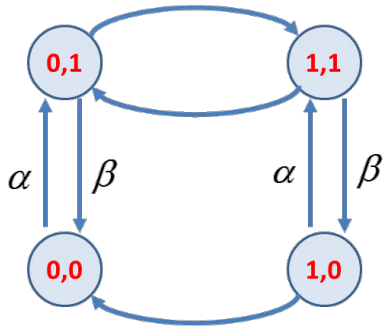


Figure 3.1

4.

a) Consider the computer system, shown in Fig. 4.1., composed of two identical sub-systems each containing two computers working in micro-synchronism.

The two sub-systems are connected to a master computer via two identical and independent buses.

If any computer of a sub-system fails, the sub-system automatically disconnect itself from the buses.

If any of the two buses fail the system will not be able to operate (as the master computer will not be able to independently, validate the information processed by the two sub-systems).

The failure rate of a computer is λ_C.

The failure rate of a bus is λ_B.

i) Define the state space of the system. [2]

ii) Derive the Markov chain representation of the system, explicitly identifying and including all the rates of transitions between states. [3]

iii) Discuss how you would modify the Markov chain representation derived in ii) to include:

a) a replacement of a sub-system as soon as any one computer fails, and

b) a replacement of the complete computer system if it becomes non-operational. [5]

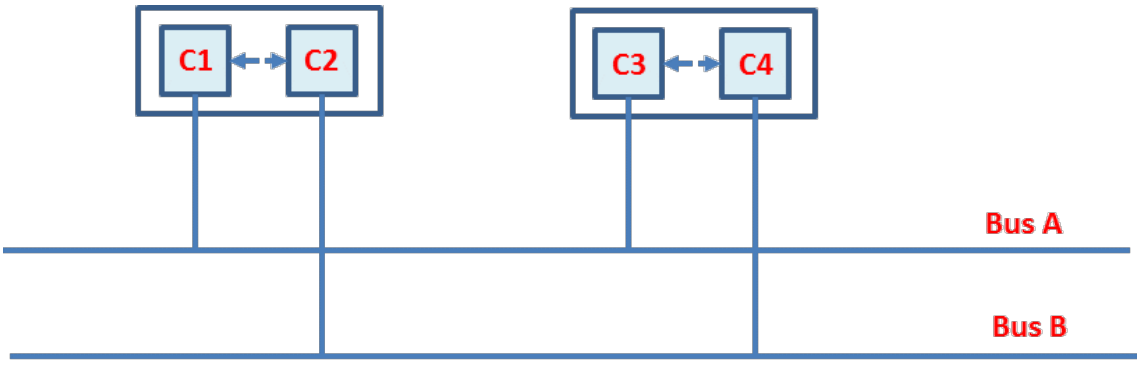


Figure 4.1

4.

b) The fluid flow model approximation of a multiplexer can be obtained by solving the following matrix equation:

$$\frac{dF(x)}{dx} D = F(x)M$$

with

$$D = diag\,[-C\alpha,\ (1-C)\alpha,\ ...,\ (N-C)\alpha]$$

$$M = \begin{bmatrix} -N\lambda & N\lambda & 0 & ... \\ \alpha & -[\alpha + (N-1)\lambda] & (N-1)\lambda & ... \\ 0 & 2\alpha & -[2\alpha + (N-2)\lambda] & ... \\ ... & ... & ... & \end{bmatrix}.$$

i) Describe in your own words the physical meaning of the parameters: C, α, N, λ in the above expressions.

Clearly identify the parameters associated with the underlying source model. [4]

ii) For a $N-$source multiplexer: derive a simple relationship between its normalised capacity and the normalised average incoming packet rate, for the multiplexer to operate in a stable regime.

Hint: You can use a simplified multiplexer with $N = 1$, followed by a discussion on its extension to a $N-$source multiplexer. [6]